



ARTIFICIAL INTELLIGENCE SELF-LEARNING CLASSICAL KHON DANCE

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Abstract:

This research extends our previous work, "Self-practice Khon Application using Artificial Intelligence," which was presented at the 2024 International Conference on Culture Technology. In this expanded study, we integrate a TimeDistributed layer to the model, introduce Dynamic Time Warping (DTW) for pose comparison, and enhance preprocessing by prioritizing joint ordering. Khon is a renowned traditional dance in South Asia and Southeast Asia, particularly in Thailand. Mastering Khon requires years of continuous training, and the limited number of Khon institutions and specialists in Thailand can make self-directed learning challenging. Notwithstanding these drawbacks, people's desire to learn about and interact with Khon has grown, which promotes appreciation and aids in the preservation of this significant cultural legacy. To address this, we develop an AI virtual instructor to enhance Khon practice. Five steps exist for constructing an artificial intelligence Khon instructor: Data collection encompassed 25 postures of the Khon dance, with 40 videos for each position, sourced from the internet and newly recorded footage. Extract body node coordinates using MediaPipe Pose from video frames, reordered by mean priority of each node from all poses, and categorize them into three datasets for three AI models according to the categories of Khon dance postures: the first model for fundamental motions, the second model for dance vocabulary, and the third model for sign language. Each dataset was partitioned into training (80%) and testing (20%) subsets. We employed Long Short-Term Memory (LSTM) in the AI model layer with the TimeDistributed layer for training AI models. 4) Assess models according to accuracy rate and F1-score. 5) Compute the significant angles of each essential postural action to compare with the correct angle using Dynamic Time Warping. We have created a web application that provides instructional videos, training recording, and performance review with accuracy scores. The computer analyzes the video clips and subsequently identifies areas for enhancement at each stage, accompanied by a percentage-based accuracy score. The application was assessed by testing with a sample group of Khon and Thai dance rally students from Suankularb Wittayalai Rangsit School, who possessed minimal understanding of Khon. The models' performance reflected the average loss during training, accuracy, and F1-score. The training results indicate mean loss values of model1, model2, and model3 as 0.00472, 0.02213, and 0.09948, respectively; accuracy values as 0.837204, 0.792031, and 0.720918; and F1-scores as 0.840297, 0.801762, and 0.789302. The sample group, consisting of Khon and Thai dance rally students from Suankularb Wittayalai Rangsit School, was tested. The most precise dancing postures for already trained students and never trained students had mean accuracies of Ten Sao at 97.24% and Tob Khao at 84.48%. The conclusive evaluation is user testing; the most precise dancing postures are Tob Khao at 88.26%, followed by Ten Sao at 84.02%, and Iang Srisa at 83.82%.